

Derek Wang

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TECHNICAL SKILLS

Languages: C++, Java, Python, JavaScript, TypeScript, SQL

Technologies: React, FastAPI, Node.js, Next.js, Express, ROS2, PyTorch, OpenCV

Engineering Tools: Git, Fusion 360, Docker, Foxglove, PostgreSQL, MongoDB, Supabase

Certifications: Docker Foundations Professional

EXPERIENCE

Autonomous Self Driving Software Engineer

Sept 2025 – Present

WATonomous

Waterloo, ON, Canada

- Implemented ROS2 navigation stack (costmaps, A* planning, Pure Pursuit) enabling autonomous rover path following in dynamic environments.
- Led YOLO11s model retraining, improving detection mAP from 30% to 80% across 3,448 labeled images via hyperparameter tuning and augmentation redesign
- Optimized YOLO11s inference to 30 FPS on rover hardware via ONNX export and onnxslim optimization.

Autonomy Developer

Aug 2025 – Feb 2026

Waterloo Aerial Robotics Group

Waterloo, ON, Canada

- Built Python hardware-in-the-loop UAV simulation using MAVLink for the 2026 UAS Student Competition.
- Eliminated GPS teleportation bugs and achieved smooth trajectory interpolation via periodic position updates

Robotics Software Programmer – VEX Robotics Competition

Sept 2024 – June 2025

Checkmate Robotics Club

Markham, Ontario

- Reduced autonomous robot positioning error by 40% by implementing odometry and PID trajectory control.
- Qualified for 2025 VEX World Championships (top 10% globally) and placed 5th/80 at Provincials.

PROJECTS

Silhouette | *Node.js, Next.js, Express, MongoDB, Gemini, Overshoot.ai SDK*

- Built a real-time AI fashion advisor with live camera input at NexHacks, earning a VC referral to continue development for Merian Brothers
- Built full-stack application to deliver real-time outfit analysis and product recommendations.
- Integrated Overshoot.ai and Gemini for live video outfit analysis, handling async inference and rate limiting to return recommendations in under 2s

Hierarchical Reasoning Retrieval | *Node.js, Express, SQLite, Gemini*

- Built an LLM-powered document retrieval system that searches structured documents (contracts, specs) using beam search over heading-based tree structures instead of flat vector similarity
- Implemented a markdown ingestion pipeline with heading-based chunking, tree construction, LLM summarization, and hybrid search combining FTS5 keyword filtering with LLM branch selection

Coffee Chat Scheduler | *React, Python, FastAPI, PostgreSQL, Google Calendar API*

- Built a full-stack booking app with React and FastAPI, integrating Google Calendar via OAuth 2.0 for real-time availability, scheduling 20+ coffee chats
- Stored booking data in PostgreSQL via SQLAlchemy, with automated event creation and availability checking

Autonomous Vehicle Control System | *C++, ROS2, Docker, CMake, Foxglove*

- Built a C++ LiDAR perception pipeline generating real-time occupancy costmaps at 0.1m resolution in Foxglove
- Developed and tested a full ROS2 autonomous navigation system with Docker, validating obstacle avoidance across 50+ simulated scenarios

EDUCATION

University of Waterloo

Expected Apr 2030

Bachelor of Applied Science in Computer Engineering (Co-op)

GPA: 3.7

- Relevant Coursework:** Linear Algebra, Fundamentals of Programming